

# Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: HOPE | Context: US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Annotation was performed by three linguists with calibration and an expert-therapist moderator [Q45, Q46, Q47], with mental health professional consultation during guideline preparation [Q95]. Aggregate Cohen's Kappa = 0.7234 (substantial) [Q49] is reasonable. The user judges the convergent-validity risk low because dialogue-act labels are communicative rather than clinical (elicitation A4). However, there are real concerns: no per-label or per-session-type Kappa breakdown is reported, so agreement on overloaded labels (ACK, GC) is unknown; no documented South Asian, peer support, or crisis worker participation in annotation; annotator demographics are limited to age and experience range [Q50] with no geographic or cultural background reported; authors acknowledge 'annotator's bias cannot be ruled out completely' [Q96]. Given the user's MODERATE priority on OC and the partial documentation, a score of 3 reflects substantial agreement on a US-clinical-centric annotation framework with unverified cross-cultural and cross-session-type calibration.

| ID  | Evidence                                                                                                                                                                                 |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q45 | "We employed three annotators who are experts in linguistics." (p.4)                                                                                                                     |
| Q46 | "To ensure the understanding of the tasks and annotation scheme, we took a sample of the dataset and asked each annotator to annotate them as per the prepared set of guidelines." (p.4) |
| Q47 | "Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency." (p.4)                                     |
| Q49 | "We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category." (p.4)                                                                          |
| Q50 | "Annotators are in the age group of 25-35, with 2-10 years of professional experience." (p.4)                                                                                            |
| Q95 | "Moreover, we consulted with mental-health professionals and linguists in preparing the annotation guideline." (p.8)                                                                     |
| Q96 | "However, the annotator's bias cannot be ruled out completely." (p.8)                                                                                                                    |

## Strengths

- Multi-annotator process with calibration exercise and expert-therapist moderation [Q46, Q47]
- Substantial aggregate inter-annotator agreement (Kappa = 0.7234) [Q49]
- Mental health professional and linguist consultation during guideline preparation [Q95]
- Authors transparently acknowledge residual annotator bias risk [Q96]

| ID  | Evidence                                                                                                                                                                                 |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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## Checklist

## Checklist item definitions:

| ID   | Assessment Question                                               |
|------|-------------------------------------------------------------------|
| OC-1 | Do ground-truth labels reflect regional stakeholder perspectives? |
| OC-2 | Assess potential annotator-regional population disagreement       |
| OC-3 | Review annotator demographics from documentation                  |
| OC-4 | Consider label re-annotation by regional annotators               |
| OC-5 | Review aggregation methods for minority perspective erasure       |
| OC-6 | Document label issues harming convergent/external validity        |

## Checklist responses:

**OC-1:** Partially — labels reflect US clinical communication norms as encoded by linguistically-trained annotators with therapist moderation; South Asian and peer/crisis stakeholder perspectives are not documented as represented.

**OC-2:** Risk is bounded for in-scope clinical structured therapy (user judges A4 risk low) but unverified for South Asian clinical contexts, peer support practitioners, and crisis workers, none of whom are documented as annotators.

**OC-3:** Limited annotator documentation [Q50] — three linguists, ages 25–35, 2–10 years experience, expert-therapist moderator. No geographic, cultural, gender, or clinical-affiliation breakdown.

**OC-4:** Re-annotation by representative regional pool (South Asian clinicians, peer support workers, crisis staff) is not documented. The user's elicitation reasoning (A4) explicitly accepts this on grounds that labels are communicative, not clinical.

**OC-5:** Aggregation method is consensus discussion moderated by expert therapist [Q47]; this could erase minority annotator perspectives (e.g., on ambiguous ACK boundaries), though no minority-perspective tracking is reported.

**OC-6:** Documented issues: aggregate-only Kappa with no per-label or per-session-type breakdown; no cross-cultural annotator validation; complementary label pairs explicitly known to be challenging [Q51, Q52]; explicit acknowledgment of unresolved annotator bias [Q96].

| ID  | Evidence                                                                                                                                                             |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| Q47 | "Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency." (p.4)                 |
| Q49 | "We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category." (p.4)                                                      |
| Q50 | "Annotators are in the age group of 25-35, with 2-10 years of professional experience." (p.4)                                                                        |
| Q51 | "The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature." (p.4) |
| Q52 | "However, the dataset does not support the above view entirely." (p.4)                                                                                               |
| Q95 | "Moreover, we consulted with mental-health professionals and linguists in preparing the annotation guideline." (p.8)                                                 |
| Q96 | "However, the annotator's bias cannot be ruled out completely." (p.8)                                                                                                |

## Information Gaps

- Per-label and per-session-type Kappa breakdown [deferred — not in available paper sections]
- Annotator geographic, cultural, and clinical-background details
- Whether South Asian clinicians or peer/crisis workers were consulted at any stage

## Requires Expert Verification

- Whether South Asian psycholinguists would label ambiguous ACK/GC utterances consistently with HOPE annotators
  - Whether peer support practitioners would assign the same labels to peer-register utterances
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## Remediation

**Gap 1: No documented South Asian, peer-support, or crisis-worker annotators; aggregate-only Kappa of 0.7234 [Q49] cannot reveal cross-cultural or cross-session-type calibration.**

**Recommendation:** Convene a small expert panel of South Asian clinical psychologists, peer-support practitioners, and crisis-intervention staff to re-annotate a stratified evaluation subset. Compute disagreement rates relative to HOPE's labels and use these to bound deployment confidence by stakeholder type.

| ID  | Evidence                                                                                                        |
|-----|-----------------------------------------------------------------------------------------------------------------|
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